

Hrigved Suryawanshi

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📁 Portfolio

🌐 LinkedIn

EDUCATION

Northeastern University, Boston, MA

Sep 2023 - May 2025

Master of Science in Robotics (with Thesis)

Relevant Coursework: Pattern Recognition, Computer Vision, Mobile Robotics, Sensing & Navigation, Robot Mechanics

University of Mumbai, Mumbai, India

Aug 2019 - May 2023

Bachelor of Engineering in Electronics and Telecommunications

Honors: Best Innovator Award for Sustainable [Electric Motorcycle](#) Project

PROFESSIONAL EXPERIENCE

Silicon Synapse Lab | Research Assistant | Boston, MA

Jun 2024 - Present

M4 (Multimodal Morphing Robot):

- Developed **ROS2** decision node leveraging CNN traversability estimator for optimal ground/aerial mode selection, integrating terrain roughness, slope, and battery efficiency into custom **NAV2** costmaps (89% accuracy); engineered elevation mapping fusing Livox LiDAR with RealSense D455 (0.05m resolution, 100Hz)
- Designed NAV2/MAVROS integration with DWB local planner, enabling 6-second morphing transitions based on traversability scores.

Cobra (Snake Robot for Martian Terrain):

- Benchmarked **SLAM** algorithms (LIO-SAM vs RTAB-Map), achieving 2cm localization accuracy validated against OptiTrack; reduced mapping error by 25% through sensor calibration
- Performed point cloud segmentation using **PCL** for terrain analysis and CloudCompare for temporal change detection, processing 100k+ points/frame to identify navigable surfaces with 94% accuracy

FYNDME | Deep Learning Engineer Intern | Boston, MA

Jan 2025 - May 2025

- Improved face recognition accuracy from 60% to 85% by implementing dual-layer anti-spoofing with **InsightFace** and DeepPixBIS, reducing false acceptance rate to 2% for secure authentication
- Achieved 92% mAP@0.5 for real-time SKU detection using **YOLOv8** across 500+ product categories, decreasing product search time for delivery drivers navigating unfamiliar retail environments

VHIRON | Embedded Systems Engineer | Mumbai, India

Oct 2022 - Mar 2023

- Developed 3D photogrammetry scanner achieving 0.1mm resolution using Raspberry Pi and motorized turntable, processing 1M+ point clouds through cloud pipeline with 50% time reduction
- Constructed 18-inch CNC laser cutter with 0.05mm precision using GRBL firmware and ran G-code optimization for rapid prototyping applications

TECHNICAL SKILLS

Programming: Python, C/C++, Bash, MATLAB, Linux

Robotics: ROS/ROS2, Gazebo, MoveIt, NAV2, MAVROS, PCL, SLAM (ORB-SLAM3, RTAB-Map, LIO-SAM, FAST-LIO)

ML/CV: PyTorch, TensorFlow, OpenCV, YOLO, MobileNet, InsightFace, DeepPixBIS, CLIP

Hardware: Jetson, Pixhawk, LiDAR, RealSense, RPi, Arduino

Tools: Docker, Git, MeshLab, CloudCompare, Solidworks, MATLAB Simulink, Colmap, PyBullet, MuJoCo

PROJECTS

Autonomous AI Drone Platform

May 2024 - Present

- Architected fully autonomous UAV with **Pixhawk PX4** and Jetson Orin, implementing Extended Kalman Filter for 9-axis IMU sensor fusion (gyro, accel, mag) with barometer achieving sub-degree orientation and RTK-GPS with 10cm precision
- Integrated Orbbec Gemini 335L depth camera for obstacle avoidance (2m clearance, 100ms reaction) and custom PID-based human tracking maintaining 3m following distance with 0.5m position accuracy
- Developed ArUco-based precision landing achieving 5cm accuracy and deployed **RTAB-Map** SLAM for real-time 3D mapping

Autonomous warehouse Mobile Robot

Aug 2024 - Present

- Deployed **FAST-LIO** SLAM on custom AMR using Livox Mid-360 on Jetson Orin, achieving 2cm odometry drift over 100m trajectories with real-time loop closure
- Implemented NAV2 navigation stack with DWB local planner and SMAC hybrid-A* global planner, enabling 2m/s navigation with dynamic obstacle avoidance in cluttered environments

Advanced Vehicle Safety System

Apr 2024 - May 2024

- Trained YOLOv8 on 4500 annotated traffic signs achieving 96% detection accuracy; integrated ByteTrack for multi-vehicle tracking at 30 FPS
- Deployed Tesseract OCR pipeline for license plate extraction with 89% accuracy, processing HD video streams in real-time

PUBLICATIONS

- Simple User-Friendly MIDI Controller using Arduino**
- ECG-based Arrhythmia Detection using CNN**